

Homework Flow — Cost Verification & Provider Reconciliation

Purpose. This document lets us *confirm* the per-session cost the estimator shows for the Homework Help (Tier 2) flow. It checks every provider unit price against the live provider documentation, then shows exactly how the biggest line — **TTS audio** — is counted, how image generation is now costed at **one diagram per session**, and explains the o3 reasoning-token risk.

Bottom line up front: the unit prices are *correct and current*. The headline number is driven by two deliberate, conservative choices: **audio is billed on the MAXIMUM narration length** (Section 2) and every price carries a **+15% buffer**. Image generation is now a single diagram per session (Section 3), so it is a minor line. The figure to keep validating on a real invoice is o3's **hidden reasoning tokens** (Section 4.3).

1. Provider unit prices — confirmed against documentation (July 2026)

Service	Model	Published list price	In estimator	Confirmed
Reasoning LLM	o3	\$2.00 / 1M input · \$8.00 / 1M output	\$2 / \$8	✓ matches OpenAI
Prose / vision LLM	gpt-4.1	\$2.00 / 1M input · \$8.00 / 1M output	\$2 / \$8	✓ matches OpenAI
Diagram image	Gemini 3.1 Flash Image	\$0.067 / image (standard tier)	\$0.067	✓ as configured
Text-to-speech	Inworld TTS-2	\$25.00 / 1M characters (on-demand)	\$25	✓ matches Inworld
Safety	omni-moderation	Free	\$0	✓

The +15% safety buffer. Every price above is multiplied by **1.15** inside the estimator before any figure is shown, as an internal margin for retries, prompt drift and future price moves. So the effective rates the estimator charges against are: o3 / gpt-4.1 **\$2.30 / \$9.20** per 1M, Gemini **\$0.077**

per image, Inworld **\$28.75** per 1M characters. Throughout, we show **both** columns — "*list*" (the provider's published price, for reconciliation) and "+ 15%" (what the estimator displays).

2. Text-to-speech (Inworld TTS-2) — billed on the MAXIMUM narration length

Inworld bills **per character** sent to synthesis. Real homework narration runs far longer than a "typical" estimate, so the estimator costs audio on the **maximum** a session can produce — every audio driver at its ceiling. This is the figure shown on the dashboard, and it is now the **single largest line in a session**.

Phase narrated	Max characters	At the slider ceiling
3A explanation	800	<code>charsExp</code> max
Solution + parallel-demo steps	10,000	<code>10 steps × 500 × 2</code>
Intros + completion	1,200	<code>chars0v</code> max
Attempt feedback + Q&A	12,000	<code>(12 + 8) × 600</code>
Maximum per session (billed)	24,000	

Cost of 24,000 characters: list `24,000 ÷ 1,000,000 × $25` = **\$0.60** · +15% = **\$0.69**.

Inworld gets cheaper with commitment, and we estimate on the most expensive on-demand tier, so a real audio invoice can only come in lower:

Inworld TTS-2 tier	\$ / 1M characters	Cost of a 24,000-char session
On-demand (what we estimate on)	\$25.00	\$0.60
\$300 / month commit	\$15.00	\$0.36
\$1,500 / month commit	\$12.50	\$0.30
Enterprise	\$10.00 (as low as \$5)	\$0.24 (↓ \$0.12)

3. Image generation (Gemini 3.1 Flash Image) — one diagram per session

Homework Help generates **exactly one diagram image per session**, regardless of walkthrough depth or how the question arrived. This is homework-only behaviour: a single supporting diagram is produced for the session and reused across the walkthrough steps rather than re-generated at each phase.

```
images / session = 1    (fixed – one diagram per session)
```

Because it no longer scales with walkthrough depth, this line is flat and small:

Images / session	Cost (list @ \$0.067)	Cost (+15%)
1 (fixed)	\$0.067	\$0.077

This is now the smallest provider line in a session. At one image, generation is **\$0.067 list / \$0.077 buffered** — well below audio and the o3 reasoning cost. If in future the flow needs more than one diagram per session, raise the image assumption in the estimator and this line scales linearly.

4. o3 reasoning model — tokens and cost

o3 is only used for **calculation / derivation** questions (Maths, Physics, Chemistry, computational Science). Prose/explanation questions run on gpt-4.1. The build pipeline is five LLM calls (steps A · parallel B · demo C · 3A explanation · answer key D); A→B→C can regenerate on a self-check failure, so they carry a retry factor (default **1.15×**).

4.1 Per-call token assumptions (o3 route)

Output tokens below are set to **include** o3's hidden reasoning tokens — see Section 4.3.

Call	Input tok	o3 output tok	Runs / session	o3 cost (list)
A · Solution steps	700	3,000	1.15	\$0.0292
B · Parallel question	800	2,500	1.15	\$0.0248
C · Parallel demo (worked)	900	3,000	1.15	\$0.0297
3A · Explanation	600	2,200	1.00	\$0.0188
D · Private answer key	900	3,000	1.00	\$0.0258
Validate attempt	700	1,500	3.00	\$0.0402
Follow-up Q&A	800	1,500	1.00	\$0.0136
o3 subtotal	~7,160	~20,975		\$0.1821

Plus the fixed gpt-4.1 intake/classify calls that run in **every** session (image extract + analyze + match): **\$0.0118**.

4.2 Per-session o3 cost (the math)

o3 input : 7,160 tok ÷ 1,000,000 × \$2 = \$0.0143
 o3 output: 20,975 tok ÷ 1,000,000 × \$8 = \$0.1678
 o3 total (list) = \$0.1821 → +15% = \$0.2094

4.3 Why o3 can look cheaper than the docs — hidden reasoning tokens

The unit price is right (\$2/\$8); the risk is the token count, not the rate. o3 bills *invisible* reasoning tokens at the **output** rate. OpenAI's documentation notes a single o3 answer can burn **5,000–20,000 internal reasoning tokens**. Our model already assumes ~21,000 o3 output tokens for a whole calculation session (near the high end), but if o3 reasons *harder*, cost scales. This table shows a full calculation session (including the max-audio \$0.60 and the one-image \$0.067 lines) as reasoning load rises:

o3 reasoning load	o3 output tok / session	Calc session (list)	Calc session (+15%)
As modelled	~21,000	\$0.861	\$0.990
1.5× harder	~31,500	\$0.945	\$1.087
2× harder	~42,000	\$1.029	\$1.183
3× harder	~63,000	\$1.197	\$1.377

The o3 output-token counts are **editable** in the estimator's *Advanced assumptions* — tune them to a sample of real billed sessions and the estimate tracks reality.

5. Full homework-session confirmation table

Per-session cost by service, with **audio at the 24,000-char maximum** and **one real-time image**. **Calculation** sessions use o3; **explanation** sessions use gpt-4.1; the **blended** row is the estimator default (50/50, image intake, audio on).

Session type	o3	gpt-4.1	Gemini image (×1)	Inworld audio (max)	Total (list)	Total (+15%)
Calculation (o3)	\$0.182	\$0.012	\$0.067	\$0.600	\$0.861	\$0.990
Explanation (gpt-4.1)	—	\$0.070	\$0.067	\$0.600	\$0.737	\$0.848
Blended (default)	\$0.091	\$0.041	\$0.067	\$0.600	\$0.799	\$0.919

Every number is reproducible from Sections 2–4 with the published unit prices — that is the confirmation: **provider price × quantity = the estimator's figure**. Audio alone is now ~70–80% of a session, because it is costed at its conservative maximum; the single image line is minor.

6. Scaling to volume (monthly / yearly)

Using the **blended buffered** cost of **\$0.919** per session, scaled by each preset's monthly session count (the *range* uses the explanation-only and calculation-only bounds, \$0.848 → \$0.990):

Preset	Students × questions	Sessions / month	Monthly (blended)	Yearly (blended)	Monthly range
Reset to defaults	200 × 10	2,000	\$1,838	\$22,056	\$1,696 – \$1,980
Light usage	300 × 20	6,000	\$5,514	\$66,168	\$5,088 – \$5,940
Heavy usage	600 × 40	24,000	\$22,056	\$264,672	\$20,352 – \$23,760

Monthly figures use the blended-default per-session cost. The two levers that move these most: the Inworld tier (a \$1,500/mo commit more than halves the audio line, the dominant cost) and the o3 reasoning-token count (Section 4.3).

7. Reconciliation summary

1. **Unit prices are correct and current** (verified July 2026) — o3 & gpt-4.1 at \$2/\$8 per 1M tokens, Gemini image at \$0.067, Inworld TTS-2 at \$25 per 1M characters on-demand.
2. **Audio is now the largest line** — billed at the maximum narration length (24,000 chars ≈ \$0.60 list / \$0.69 buffered).
3. **Image generation is a single diagram per session** — one Gemini image ≈ **\$0.067 list / \$0.077 buffered**, a minor line that no longer scales with walkthrough depth.
4. **Everything carries a +15% buffer** and audio is priced on Inworld's most expensive on-demand tier, so the estimate is deliberately conservative.
5. **Validate two things on real usage:** (a) the **maximum narrated characters** per session vs. the 24,000-char ceiling — the dominant line; and (b) o3's reasoning-token count (Section 4.3).

Recommended checks before committing to a budget

- Confirm the **maximum narrated characters** per session vs. the 24,000-char ceiling.
 - Pull the **actual o3 output-token count** from a dozen real calculation sessions vs. ~21,000/session.
 - Decide the Inworld **commitment tier** (on-demand \$25 → \$1,500/mo commit \$12.50 halves audio).
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Sources

- OpenAI o3 API pricing (\$2 / \$8 per 1M tokens; hidden reasoning tokens billed at output rate): <https://developers.openai.com/api/docs/pricing> · <https://pricepertoken.com/pricing-page/model/openai-o3>
- Inworld TTS-2 pricing (\$25 / 1M characters on-demand, tiered to \$10 enterprise): <https://inworld.ai/pricing> · <https://www.buildmvpfast.com/api-costs/ai-voice>

Figures are grounded in the estimator's editable token / character / image assumptions; tune those to your real provider contracts and billed usage.